

ACADEMIC CITY UNIVERSITY

Machine Learning (CS/IT4232-RE3228)

Mid-Semester

March 2026

Project Requirements

Dataset Access: Download the specified dataset from public sources like Kaggle or UCI Machine Learning Repository. Ensure ethical use and proper citation.

Tasks:

- Perform exploratory data analysis: Data cleaning (handle missing values, outliers, categorical encoding), feature scaling, and create visualizations (e.g., scatter plots, histograms, confusion matrices, ROC curves, learning curves).
- Implement machine learning models: Train and evaluate models using techniques from the course (e.g., linear/polynomial regression, logistic/softmax regression, SVM with kernels, decision trees, ensembles).
- Evaluate performance: Use metrics like accuracy, precision, recall, F1 score, MSE, confusion matrices, precision/recall trade-off, ROC curves; detect overfitting/underfitting with learning curves; balance bias/variance via regularization.
- Compare models: Experiment with hyperparameters (e.g., learning rate, C in SVM, gamma in RBF kernel, max_depth in trees) and discuss results.

Deliverables:

- Jupyter Notebook script with code, explanations, and results.
- Final PowerPoint presentation (10-15 slides): Include project title, problem description, dataset overview, EDA visualizations, model implementations, evaluation metrics, insights, challenges, and recommendations. Present to the class.

Group Assignments

Group 1 - Predicting Diabetes Risk in Patients Dataset: Pima Indians Diabetes Database from UCI Machine Learning Repository (Kaggle).

Problem Description: Help healthcare providers identify patients at risk of developing diabetes based on medical attributes like glucose levels and BMI, enabling early intervention.

Tasks:

- Perform data cleaning to handle missing values; create scatter plots to visualize relationships between features like insulin and glucose.
- Train a logistic regression model for binary classification (diabetes vs. no diabetes).
- Implement a support vector machine (SVM) with soft margins and RBF kernel.
- Build a decision tree classifier; evaluate models using confusion matrices, precision, recall, F1 score, and ROC curves; Detect overfitting/underfitting with learning curves; apply ensemble methods like AdaBoost and voting classifiers;

Group 2 - Forecasting Basketball Player Salaries Dataset: NBA Players Stats from Kaggle (includes player performance metrics).

Problem Description: Assist sports teams in estimating player salaries based on performance metrics like points per game and assists, to inform contract negotiations.

Tasks:

- Data cleaning to remove outliers.
- Create scatter plots for features like age vs. points; implement linear regression using the normal equation and gradient descent (batch, stochastic, mini-batch).
- Apply polynomial regression for nonlinear relationships.
- Use Ridge and Lasso regularization; detect overfitting/underfitting with learning curves.

- Balance bias/variance trade-off; evaluate with mean squared error (MSE).

Group 3 - Detecting Customer Churn in Telecom Dataset: Telco Customer Churn from Kaggle. **Problem Description:** Help telecom companies predict which customers are likely to leave based on usage patterns and demographics, allowing targeted retention strategies.

Tasks:

- Data cleaning for categorical encoding
- Create scatter plots of tenure vs. monthly charges; train logistic regression for binary classification (churn vs. no churn)
- Build SVM with polynomial kernel; construct a decision tree; apply ensemble techniques like random forests and XGBoost;
- Evaluate using confusion matrices, precision, recall, F1 score, and ROC curves; analyze precision/recall trade-off.

Group 4 - Assessing Air Quality Levels Dataset: Air Quality Data in India from Kaggle.

Problem Description: Enable environmental agencies to classify air quality as safe or hazardous based on pollutants like PM2.5 and NO2, supporting public health alerts.

Tasks:

- Data cleaning to handle missing sensor data; create scatter plots of PM10 vs. ozone;
- Perform multiclass classification using softmax regression; train SVM with Gaussian RBF kernel and gamma tuning;
- Build decision trees with Gini impurity; use bagging and boosting (AdaBoost);
- Evaluate with confusion matrices, F1 score, and ROC curves (extended to multiclass); detect bias/variance with learning curves.

Group 5 - Predicting Pet Adoption Speeds Dataset: PetFinder.my Adoption Prediction from Kaggle. **Problem Description:** Assist animal shelters in forecasting how quickly pets like cats and dogs will be adopted based on attributes such as age and breed, to optimize care resources.

Tasks:

- Data cleaning for handling categorical features; create scatter plots of age vs. adoption speed;
- Implement linear regression with gradient descent variants; apply polynomial regression;
- Use Elastic Net regularization; evaluate MSE and detect overfitting with learning curves; balance bias/variance;
- For classification aspect, use logistic regression on binary outcomes (fast vs. slow adoption).

Group 6 - Identifying Credit Card Fraud Dataset: Credit Card Fraud Detection from Kaggle.

Problem Description: Aid banks in detecting fraudulent transactions based on features like amount and time, reducing financial losses.

Tasks:

- Data cleaning to address class imbalance; create scatter plots of transaction amount vs. time;
- Train logistic regression for binary classification; implement SVM with soft margins for outlier handling; build decision trees;
- Apply ensemble methods like random forests and XGBoost;
- Evaluate using confusion matrices, precision, recall, F1 score, ROC curves, and precision/recall trade-off analysis.

Group 7 - Forecasting Student Academic Performance Dataset: Student Performance from UCI Machine Learning Repository.

Problem Description: Support educators in predicting student grades based on study habits and family background, to provide timely interventions.

Tasks:

- Data cleaning for missing values; create scatter plots of study time vs. grades;
- Implement linear regression using normal equation and mini-batch gradient descent;
- Apply polynomial regression; use Ridge regularization;
- Detect underfitting/overfitting with learning curves; balance bias/variance; evaluate with MSE.

Group 8 - Classifying Traffic Sign Images Dataset: German Traffic Sign Dataset from UCI Machine Learning Repository (Kaggle).

Problem Description: Help autonomous vehicles recognize traffic signs based on image features, improving road safety.

Tasks:

- Data cleaning to normalize pixel intensities; create scatter plots (if applicable, e.g., reduced dimensions);
- Perform multiclass classification with softmax regression; train SVM with polynomial kernel;
- Build decision trees; use voting classifiers and random forests;
- Evaluate with confusion matrices, F1 score, and ROC curves (multiclass); analyze error using confusion matrices.

Group 9 - Estimating Crop Yields for Farmers Dataset: Crop Yield Prediction from Kaggle.

Problem Description: Assist farmers in predicting crop yields based on factors like rainfall and soil nutrients, optimizing planting decisions.

Tasks:

- Data cleaning for outliers; create scatter plots of rainfall vs. yield;
- Implement linear regression with stochastic gradient descent;
- Apply polynomial regression; use Lasso regularization; detect bias/variance with learning curves;
- Evaluate MSE; for binary quality classification, use logistic regression.

Group 10 - Predicting House Prices in Real Estate Dataset: Ames Housing Dataset from Kaggle. **Problem Description:** Enable real estate agents to estimate house prices based on features like size and location, aiding buyers and sellers.

Tasks:

- Data cleaning to handle categorical variables; create scatter plots of living area vs. price;
- Train linear regression using batch gradient descent;
- Implement polynomial regression;
- Apply Elastic Net regularization; evaluate with MSE; detect overfitting/underfitting using learning curves; balance bias/variance trade-off.

Group 11 - Classifying Wine Quality Ratings Dataset: Wine Quality from UCI Machine Learning Repository.

Problem Description: Help winemakers assess wine quality based on chemical properties like acidity and alcohol content, improving production standards.

Tasks:

- Data cleaning for feature scaling; create scatter plots of pH vs. quality;
- Perform multiclass classification using softmax regression; train SVM with RBF kernel and gamma; build decision trees with entropy;
- Apply boosting (XGBoost) and bagging;
- Evaluate using confusion matrices, precision, recall, F1 score, and ROC curves; analyze precision/recall trade-off.